Agentic Delegation and the Language Frontier

What the threshold model predicts—and what the design cannot identify

Gabriel Saco

August 29, 2026

Universidad del Pacífico

Quispe & Xu (2026) · arXiv:2605.25438

github.com/gsaco/ai-03-quispe/tree/branch1

The paper and the developer's problem

Question and economic mechanism

Question. Does an agent expand the set of programming languages in which a developer can ship code? This is a production frontier, not a claim that the developer learns each language.

Mechanism. Conversational AI augments existing language skill; agentic AI adds delegated execution under human specification and verification.

Choice problem for developer i, language k, month t

Pre-agent menu $\mathcal{M}_1 = \{S, C\}$; post-agent menu $\mathcal{M}_2 = \{S, C, D\}$:

$$m^*_{ikt} \in \arg\max_{m \in \mathcal{M}_g} V^m_{ikt}, \qquad Z^g_{ikt} = \mathbb{I}(\max_m V^m_{ikt} \geq 0).$$

Monthly breadth is $N^g_{it} = \sum_k Z^g_{ikt}$.

S**:** own execution.

C**:** suggestion gain $\gamma s - r_C$.

D**:** agent execution $\lambda a z(A)$, net of verification $\kappa(a, s)$, compute r_D, and residual risk.

Main result: a conditional activation band

Proposition 2 for an unfamiliar language

If conversational assistance has no value without a skill foothold, $\gamma \underline{s} - r_C \leq 0$, then $T^1 = T^S$. If delegation lowers the threshold, $B \equiv T^S - T^D > 0$, then

$$Z^2 - Z^1 = \mathbb{I}(T^D \leq \omega < T^S), \qquad \Pr(Z^2 - Z^1 = 1) = F(T^S) - F(T^D).$$

Conditions that do the work

Theory

- no conversational foothold in k;
- verification, compute, and residual-risk costs leave $B > 0$;
- opportunity mass lies inside $[T^D, T^S)$;
- the developer can specify and verify.

Empirical interpretation

- public, sustained adopters; "new" since Jan. 2024;
- not-yet adopters are valid counterfactuals;
- no time-varying project shock jointly moves adoption and languages.

Observed at adoption: +2.53 active languages and +1.19 newly used languages; these are event-time associations, not identified causal effects.

What I did: two pressure tests

1. The activation band does not uniquely require an agent

A generic productivity or fixed-cost improvement $\tau > 0$ gives $T^G = T^1 - \tau$ and therefore

$$Z^G - Z^1 = \mathbb{I}(T^1 - \tau \leq \omega < T^1).$$

This is the same comparative static. Agent specificity comes from how the modes are assumed, not from the threshold result itself.

2. A zero-effect selection counterexample

Let an unfamiliar-language project shock both trigger adoption and raise breadth:

$$A_{it} = \mathbb{I}[P_{it} \geq c_i], \qquad Y_{it} = \mu_i + \lambda_t + \beta P_{it} + \tau A_{it} + \varepsilon_{it}.$$

At first adoption,

$$\widehat{\mathrm{ATT}}(e) = \tau + \beta[\mathbb{E}(P_{it} \mid G_i = g) - \mathbb{E}(P_{it} \mid G_i > t)]\,.$$

The repository simulation sets $\tau = 0$ and still produces a sharp event-time jump.

The robustness checks clean the outcome; they do not exogenize the timing of adoption.

Hand check: where I did not believe the AI

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Add `hand/selection-derivation.jpg`

Copy the four-line derivation in `hand/README.md`.

The AI claim I challenged

Flat pre-trends and outcome-side robustness make the portfolio expansion persuasive evidence for agentic delegation.

What the derivation shows

With $\tau = 0$,

$$\widehat{\mathrm{ATT}}(0) = \beta\{\mathbb{E}[P \mid P \geq c] - \mathbb{E}[P \mid P < c]\} > 0$$

whenever $\beta > 0$. The common project shock creates adoption and diversification together. Earlier pre-trends may still be flat because the shock arrives at $e = 0$.

Verdict: mechanism-consistent, not mechanism-identified

The paper documents a large, robust coincidence but cannot distinguish delegation from project selection. Causal identification needs exogenous adoption variation or a comparison with equally large non-agent productivity shocks.